

TED UNIVERSITY

Department of Computer Engineering

CMPE 491 – Senior Design Project I

TODO / BACKLOG DOCUMENT

D3 · Version 1.0

Improving Object Detection Performance in SAR Imagery Using Super-Resolution Techniques

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Fall 2026 · Group 5 · 23 October 2026

1. DOCUMENT INFORMATION

ITEM	DETAILS
Project	Improving Object Detection Performance in SAR Imagery Using Super-Resolution Techniques
Team	KESKİN (Group 5): Busenur Arlı, İlke Özçendek, Zeynep Ersoy
Supervisor	Prof. Dr. Gökçe Nur Yılmaz
Jury Members	Dr. Mert Burkay Çöteli, Prof. Dr. Tolga Kurtuluş Çapın
Deliverable	D3 – TODO / Backlog Document, Version 1.0
Date	23 October 2026
Next Updates	Version 2.0 (D5) on 20 November 2026; Version 3.0 (D7) on 31 December 2026. Each version is published on the project website and evaluated by the supervisor.
Project Website	https://ilkeozcendek.github.io/keskin/

2. CURRENT PROJECT STATUS

The project proposal (D1) and the Project Specifications – Literature Review report (D2) have been completed and submitted. Public copies are available on the project website with identity numbers removed. The specification defines the evaluation design: five input conditions, two detector protocols (P1 and P2), and requirements EVAL-01 to EVAL-06. Three decisions remain open: the dataset (HRSID or SSDD), the two super-resolution architectures (candidates: EDSR and SRGAN), and the detector (for example, YOLOv8).

Until the Analysis Report on 20 November 2026, the team will confirm these decisions with the supervisor, run small feasibility tests of data preparation and evaluation, and write the Analysis Report. Full super-resolution training and the P1 and P2 experiments will follow once the decisions are confirmed.

3. PROJECT BACKLOG

Each task has a fixed ID. Near-term tasks have target dates; later tasks will be dated in future versions. Shared deliverables are split into sub-tasks (for example BL-16.1), and the parent task covers merging and submission. Status is To Do, In Progress, Blocked or Done. A task is marked Done when its evidence is available in the last column. Near-term tasks are planned in detail; later tasks will be refined in the next versions.

TABLE 1. *Project backlog, version 1.0.*

ID	TASK / EXPECTED OUTPUT	RELATED REQ.	PRIORITY	TARGET DATE	STATUS	DEPENDENCY / EVIDENCE
COMPLETED						
BL-01	Project proposal (D1) prepared and submitted	—	High	16 Oct 2026	**Done**	Proposal v1.0 (PDF) Public copy: 23 Oct 2026
BL-02	Project website set up and published	—	High	16 Oct 2026	**Done**	KESKİN project website
BL-03	Project Specifications – Literature Review (D2) prepared and submitted	—	High	23 Oct 2026	**Done**	Specifications v1.0 (PDF) Public copy: 23 Oct 2026
BL-04	TODO / Backlog Document v1.0 (D3) published	—	High	23 Oct 2026	**Done**	Backlog v1.0 (this PDF) Website publication: 23 Oct 2026
BL-15	Remove identity numbers from the public copies of the project reports	Spec. 1.3	High	30 Oct 2026	**Done**	Public proposal PDF Public specifications PDF Completed: 23 Oct 2026
UNTIL THE ANALYSIS REPORT · 20 NOVEMBER 2026						
BL-05	Confirm the open decisions with the supervisor: dataset, SR architectures, detector, and the EVAL-02 and EVAL-05 rules → meeting notes	Spec. 2.1, 2.4	High	30 Oct 2026	**To Do**	Meeting notes in the repository

ID	TASK / EXPECTED OUTPUT	RELATED REQ.	PRIORITY	TARGET DATE	STATUS	DEPENDENCY / EVIDENCE
BL-06	Select the dataset (HRSID or SSDD) → written rationale	DATA-05	High	30 Oct 2026	**To Do**	After BL-05
BL-07	Check access, licence and annotation format → data card	DATA-05	High	6 Nov 2026	**To Do**	After BL-06
BL-08	Create and freeze the train / validation / test splits; keep each source image and all its derivatives in the same split → split files	DATA-02, DATA-04	High	6 Nov 2026	**To Do**	After BL-07
BL-09	Set up the code repository and environment: seeds, configuration files, README outline	NFR-01, NFR-02, NFR-05	Medium	6 Nov 2026	**To Do**	Repository link
BL-10	Pilot ×2 / ×4 LR–HR generation on a small sample; verify image sizes and rescaled boxes	DATA-01, DATA-03, FR-01	High	13 Nov 2026	**To Do**	After BL-07
BL-11	Generate bicubic baseline images for the pilot sample	FR-03	Medium	13 Nov 2026	**To Do**	After BL-10
BL-12	Finalise the two SR architectures and the detector; check that they fit the available GPU → selection note	FR-02, FR-04, NFR-03	High	13 Nov 2026	**To Do**	After BL-05
BL-13	Test the detector input size and padding behaviour on pilot images → short test record	EVAL-05	Medium	13 Nov 2026	**To Do**	After BL-10, BL-12
BL-14	Prepare the evaluation scripts (PSNR / SSIM, precision / recall / mAP, size groups) and the result-table format; run them on pilot outputs	EVAL-01–04, EVAL-06, FR-05	Medium	13 Nov 2026	**To Do**	After BL-11, BL-13
BL-16	Analysis Report (D4): merge the sections, final check and submission (LMS, e-mail to supervisor and jury, coordinator form, website)	—	High	20 Nov 2026	**To Do**	After BL-16.1 – BL-16.3
BL-16.1	Analysis Report: introduction, current system, proposed system overview, glossary	—	High	17 Nov 2026	**To Do**	After BL-05
BL-16.2	Analysis Report: functional, nonfunctional and pseudo requirements; object and class model	Spec. Section 2	High	17 Nov 2026	**To Do**	After BL-06, BL-12

ID	TASK / EXPECTED OUTPUT	RELATED REQ.	PRIORITY	TARGET DATE	STATUS	DEPENDENCY / EVIDENCE
BL-16.3	Analysis Report: scenarios, use-case model, dynamic models, user-interface views	Spec. 2.4	High	17 Nov 2026	**To Do**	After BL-13, BL-14
BL-17	TODO / Backlog Document v2.0 (D5) published	—	Medium	20 Nov 2026	**To Do**	After BL-16
UNTIL THE HIGH-LEVEL DESIGN REPORT · 31 DECEMBER 2026						
BL-18	Generate the ×2 / ×4 LR–HR pairs and the bicubic images for all splits	FR-01, FR-03, DATA-01, DATA-03	Medium	To be set	**To Do**	After BL-08, BL-10, BL-11
BL-19	Train both SR architectures at ×2 and ×4 on the SAR training split; record time and memory	FR-02, NFR-03, NFR-04	Medium	To be set	**To Do**	After BL-12, BL-18
BL-20	Apply the trained SR models to the LR images of all splits	FR-06	Medium	To be set	**To Do**	After BL-19
BL-21	High-Level Design Report (D6): merge sections, final check and submission (LMS, e-mail to supervisor and jury, coordinator form, website)	—	High	31 Dec 2026	**To Do**	After BL-21.1 – BL-21.3
BL-21.1	High-Level Design: introduction, design goals, current architecture, glossary	—	High	24 Dec 2026	**To Do**	After BL-16
BL-21.2	High-Level Design: subsystem decomposition, hardware / software mapping, persistent data management	NFR-03	High	24 Dec 2026	**To Do**	After BL-16
BL-21.3	High-Level Design: global software control, boundary conditions, access control and security, subsystem services	EVAL-05	High	24 Dec 2026	**To Do**	After BL-16
BL-22	TODO / Backlog Document v3.0 (D7) published	—	Medium	31 Dec 2026	**To Do**	After BL-21

ID	TASK / EXPECTED OUTPUT	RELATED REQ.	PRIORITY	TARGET DATE	STATUS	DEPENDENCY / EVIDENCE
LATER · DATES TO BE SET						
BL-23	Run the detection experiments under P1 and P2; record time and memory	FR-04, EVAL-02–05, NFR-04	Medium	To be set	**To Do**	After BL-13, BL-14, BL-20
BL-24	Gain analysis: combine the result tables, summarise the findings and limitations	FR-05	Medium	To be set	**To Do**	After BL-24.1, BL-24.2
BL-24.1	Detection result tables per condition, scale and size group (P1 and P2)	EVAL-02–05	Medium	To be set	**To Do**	After BL-23
BL-24.2	Image-quality tables (PSNR / SSIM) for bicubic and SR outputs, with visual examples	EVAL-01, EVAL-06	Medium	To be set	**To Do**	After BL-20
BL-25	Reproducibility re-run from the documented instructions	NFR-01, NFR-05	Low	To be set	**To Do**	After BL-24
BL-26	Presentation (D8): slides and recording; every member presents their own part	—	High	4–17 Jan 2027 (date TBA)	**To Do**	After BL-21
BL-26.1	Poster (D8)	—	High	4–17 Jan 2027 (date TBA)	**To Do**	After BL-21
BL-26.2	Prototype demonstration (D8)	—	High	4–17 Jan 2027 (date TBA)	**To Do**	After BL-21

4. UPCOMING MILESTONES, DEPENDENCIES AND RISKS

4.1 Milestones

DATE	MILESTONE
30 Oct 2026	Open decisions confirmed and dataset selected (BL-05, BL-06)
13 Nov 2026	Pilot data preparation and evaluation scripts working (BL-10 – BL-14)
17 Nov 2026	Analysis Report section drafts ready for merging (BL-16.1 – BL-16.3)
20 Nov 2026	Analysis Report (D4) and Backlog v2.0 (D5)
Nov 2026	SAYZEK-ATP application results announced (external milestone)
24 Dec 2026	High-Level Design section drafts ready for merging (BL-21.1 – BL-21.3)
31 Dec 2026	High-Level Design Report (D6) and Backlog v3.0 (D7)
4–17 Jan 2027	Presentations, poster and demonstration (D8); exact date to be announced

4.2 Dependencies

1. BL-05 → BL-06, BL-12: the dataset and model choices wait for the supervisor's confirmation.
2. BL-06 → BL-07 → BL-08, BL-10: the data card, the splits and the LR–HR pilot need the selected dataset.
3. BL-10, BL-12 → BL-13: the input-size test needs pilot images and the chosen detector.
4. BL-08, BL-10 and BL-11 → BL-18 → BL-19 → BL-20 → BL-23: full data generation and training follow the frozen splits, pilot and confirmed models (BL-12). Detection experiments also depend on the input-size check (BL-13) and evaluation scripts (BL-14).

4.3 Risks

TABLE 2. *Risks and planned responses.*

RISK	EFFECT	PLANNED RESPONSE
Dataset access or licence terms are unclear	Delays BL-06 to BL-08	Keep the other candidate dataset (HRSID or SSDD) as a fallback; record the terms in the data card (DATA-05)
Limited GPU time (cloud limits or shared servers)	Super-resolution training and P1 runs take longer than planned	Test on small samples first; record time and memory (NFR-04); apply any reduced setting equally to all conditions
Objects become very small at ×4	Few or no detections in the ×4 conditions	Keep the same test targets (DATA-03) and report results per size group (EVAL-03)
SRGAN was introduced for ×4 upscaling	Extra work to run it at ×2	Decide in BL-12 and document any adaptation
Overlapping deadlines (other courses, SAYZEK reports if accepted)	Tasks slip	Review the backlog weekly and move dates openly in the next version

5. VERSION HISTORY

VERSION	DATE	CHANGES
1.0	23 Oct 2026	Initial public backlog: submitted reports and website publication recorded, identity numbers removed from public reports, detailed tasks until 20 November and later tasks outlined.